13.01.2023

Roll No. | | | | | | | |

Paper Code: **TCS 302**

END SEMESTER Examination 2023

B.Tech (CSE/CE/CST/AI_ML/AI_DS) IIISem

Data Structure with 'C' language.

Time : Three Hours

Maximum Marks :100

NSTRUCTIONS TO STUDENTS

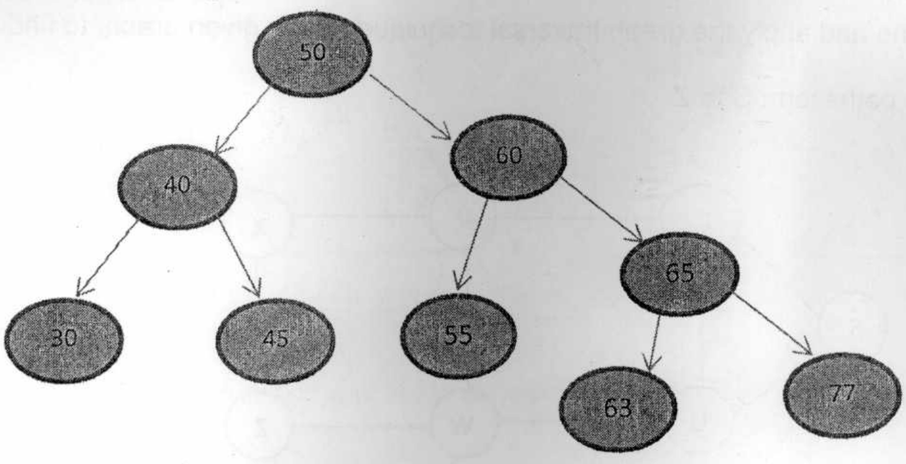
Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

Q1. (2X10=20 Marks)(CO1, CO3,CO5)

A. Consider the following tree.



i) Delete a node with info. 65 and redraw the tree

ii) Insert a node with info 62 and redraw the tree

B. *Assume that you have a singly linked list with a pointer P pointing the first node of the linked list, Write a C function to create two linked lists positive and negative from the given linked list, so that positive linked list contains all positive elements and negative linked list contains all negative elements.*

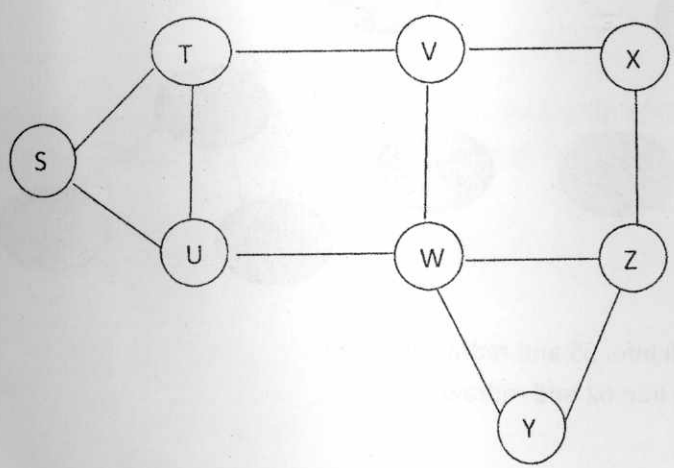
C. Draw an expression tree using following infix expression (S+T)*U + (V*W) - (X-Y) ^ Z

Q2. (2X10=20 Marks)(CO2, CO3,CO4)

A. *Write a C program to sort an unsorted sequence of strings given by user in an array, using Quick sort technique.*

B. *Assume that you have a single linked list; first node of the linked list is pointed by a pointer PTR. Write a C function to input a key value and then count nodes having information smaller than the key value.*

C. Give name and apply the graph traversal technique on the given graph, to find one of the possible paths form S to Z.



Q3. (2X10=20 Marks)(CO2, CO4,CO5)

A. *Write advantages of an AVLtree.Draw an AVL tree with following keys:*

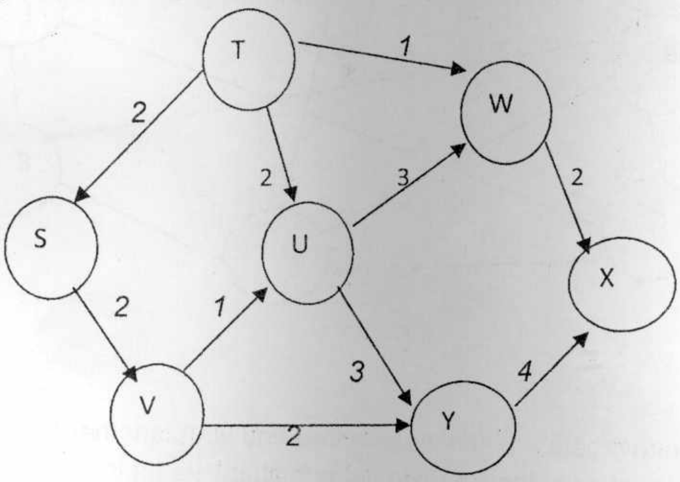
10,8,9,15,18,6,7,20,30,11,12

B. *Explain hash collision with an example. Consider a hash table of size (m) 10. Using linear probing technique insert following keys 12,17,19,22,83,21,92,55 and 311 into the hash table.*

C. Assume that you have a single linked list, with a pointer P, pointing the first node of the linked list. Write a C function to delete the second last node of the linked list.

Q4. (2X10=20 Marks)(CO1, CO3,CO5)

A. *What do you mean by minimal spanning tree? Find minimal spanning tree from the given graph using Kurskal's algorithm (show all steps).*



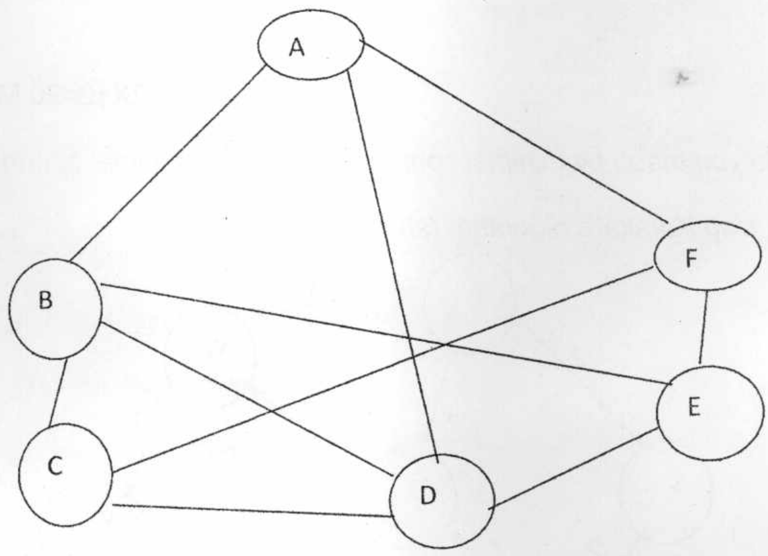
B. Explain relative file organization and index sequential file organization with examples

C. Explain big oh notation.Write an algorithm to find the sum of elements stored in two 2-D arrays, then count total numbers of steps required by the algorithm also predict the nature of the algorithm.

Q5. (2X10=20 Marks)(CO3, CO4, CO5)

A. Explain B and B^+ Tree. Draw a B tree of order 4 using the following keys,4,10,15,20,5,6,8,25,1,3,30,35,45,

B. *What do you mean by a connected graph? Give linked representation and memory representation of following graph.*



C. Write a 'C' function to create a binary search tree and write another function to print the node having highest information in the BST (do not use global variables).

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MAI–303

M. A. B. (AI&DS) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

DATA VISUALIZATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All question paper contains two Sections- Section A and B.

(ii) Both sections are compulsory.

(iii) Answer any *two* sub-questions among a, b, & c in each main questions of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 marks.

Section—A

1. (a) Why is data visualization necessary. Site real life examples. (CO1)

(b) Describe different types of data visualization. Describe the first incident in the data visualization. (CO1)

(c) What are activation functions in deep learning. Explain at east two of them. (CO1)

2. (a) What is multilayer perception and what is the threshold value ? (CO1)

P. T. O.

(b) Why was deep learning proposed when machine learning was available for data analysis. Explain two examples of deep learning applications. (CO1, 2)

(c) What type of data is analysed using deep learning architectures ? (CO1)

3. (a) Explain the overfitting and underfitting of an algorithm. (CO1)

(b) Describe the different types of layers used in deep learning architectures. (CO1)

(c) Describe different types of data used for data analysis. (CO1)

4. (a) What is the difference between a bar-chart and a histogram ? Where would you use a bar chart and where would you use a histogram ? (CO1, 2)

(b) What is gradient descent. Explain with Example. (CO1, 2)

(c) Discuss some of the important libraries in python and R that are used for visualization. (CO1)

5. **Case Study :** (20 Marks)

The data scientists at Big Mart have collected 2013 sales data for 1559 products across 10 stores in Different cities. Also, certain attributes of each product and store have been defined. The aim is to build a predictive model and find out the sales of each product at a particular store.

Using this model, Big Mart will try to understand the properties of products and stores which play a key role in increasing sales. You need to predict sales of the test dataset.

(a) What will be your approach to solve the above problem ? (CO1, 2, 3, 4)

(b) Discuss which Visualization chart you are going to use. (CO1)

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TMA–316

B. TECH. (CS/IT) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

DISCRETE STRUCTURES AND COMBINATORICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) (i) Let A = {1, 2, 3}. Give an example of a relation R in A. Such that R is neither symmetric nor antisymmetric. (CO1)

(ii) Let X = $\{a, b, c\}$. Define $f : X \to X$ such that $f = \{(a,b),(b,a),(c,c)\}$. Find :

(1) f^2

(2) f^3

(c) f^4

(b) Draw the Hasse diagram of D_{30} (positive divisors of 30) and also find its complement. (CO1)

P. T. O.

(c) If R and S are equivalence relations on a set A, show that the following are equivalence relations : (CO1)

(i) $R \cup S$

(ii) $R \cap S$

2. (a) A can hit a target 3 times in 5 shots, B 2 times in 5 shots and C 3 times in 4 shots. All of them fire one shot each simultaneously at the target. What is the probability that : (CO2)

(i) 2 shots hit

(ii) at least two shots hit ?

(b) Assuming half the population of a town consumes chocolates and that 100 investigators each take 10 individuals to see whether they are consumers, how many investigators would you expect to report that three people or less were consumers ? (CO2)

(c) Let $x_1, x_2, ..., x_n$ denote a random sample from a population with p.d.f. : (CO2)

$$f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) Write down the joint p.d.f. of $x_1, x_2, ..., x_n$.

(ii) Find the probability that the first observation is less than 0.5, $P(x_1 < 0.5)$.

(iii) Find the probability that all of the observations are less than 0.5.

3. (a) There are 4 black, 5 red and 6 green balls. We have to select 3 balls. In how many different ways we can select : (CO5)

   (i) All balls are of different colours.

   (ii) All balls are of same colour.

   (iii) No ball is red.

   (iv) At least one is a red ball.

   (v) Exactly one is green ball.

   (b) Check the validity of the argument : (CO3)

   "If I get the job and work hard, then I will get promoted. If I get promoted, then I will be happy. I will not be happy. Therefore, either I will not get the job or I will not work hard."

   (c) Use induction to shown that : (CO3)

$$n! \geq 2^{n-1} \quad \text{for } n \geq 1$$

4. (a) Show that the set G = {1, 2, 3, 4, 5, 6} forms an Abelian group with respect to multiplication modulo 7. Is it a cyclic group ? (CO4)

   (b) The order of each sub-group of a finite group G is a divisor of the order of the group G. (CO4)

   (c) Solve the recurrence relation : (CO5)

$$a_n - 7a_{n-1} + 12a_{n-2} = n.4^n$$

5. (a) Solve the following difference equation by the method of generating function $a_n - 5a_{n-1} + 6a_{n-2} = 3^n, n \geq 2$ with the boundary condition : $a_0 = 0, a_1 = 2$. (CO5)

(b) Define the following with examples : (CO6)

(i) Pseduo Graph

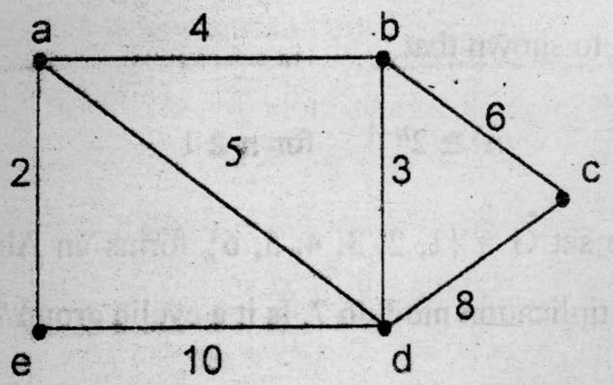
(ii) Trail

(iii) Circuit

(iv) Complete Graph

(v) Bipartite Graph

(c) Using Kruskal's algorithm, find a spanning tree with minimum weight from the graph below. Also calculate the total weight of spanning tree. (CO6)



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TCS–307

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

OBJECT-ORIENTED PROGRAMMING WITH C++

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1\. (a) (i) How data and functions are organized in object-oriented programming ? Explain with example.

(ii) Differentiate between classes and structure with example. (CO1)

(b) Write a C++ program to print duplicate characters along with number of occurrences in a given string. Characters are case sensitive. (CO1)

Input : Best of Luck StudenTs

Output : Duplicate Characters are :

e-2

s-2

t-2

u-2

P. T. O.

(c) (i) What will be output of the given code ? Justify your answer with proper explanation. (CO1)

A.

```
#include <iostream>
using namespace std;
class Data
{
    static int a;
public :
    Data()
    {
        a= 10;
    }
    void print()
    {
        cout << a<< endl;
    }
};
int main()
{
    Data obj;
    obj.print() ;
    return 0;
}
```

B.

```
#include <iostream>
using namespace std;
class ABC
{
protected :
  int a = 20;
  void show()
    {
    cout<<"a="<<a<<endl;
    }
public :
    int b= 30;
};
class XYZ : protected ABC
{
public :
}
  void disp()
  {
        cout<<"b="<< b<<endl;
  }
};
int main()
{
 XYZ obj;
 obj.show() ;
 obj.disp() ;
 return 0;
}
```

(ii) Explain how access specifiers are used for data hiding in object-oriented programming. Differentiate between different types of access specifier available in C++.

2. (a) List name of operators that cannot be overloaded in C++ programming. Write a C++ program to create a class Height having two data members foot and inch. Overload == operator using friend function to compare that two Height being denoted by the two Height type objects are similar or not. (CO2, CO3)

(b) (i) Differentiate between compile time and run time polymorphism.

(ii) How a friend function will access private data members of a class for which it is being declared as friend. Give your answer with suitable example. (CO2, CO3)

(c) Constructor program with Array of object program.

A class Shop is having following private data members :

P_id	To store id of the products
p_name	To store products name
price	To store each product price
quantity	To store available quantity of the products

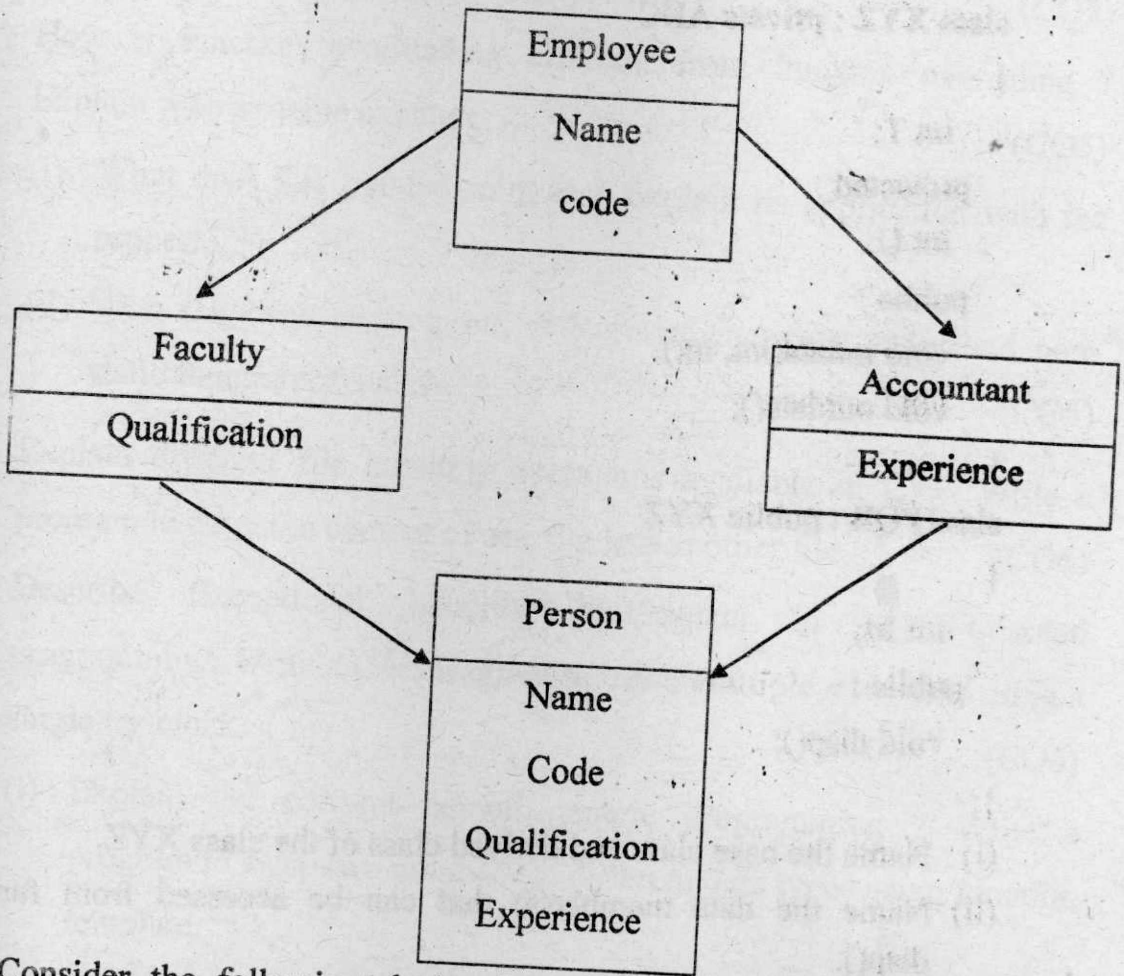
This class is having a parameterized constructor to initialize the values of data members and display these details using showdata() member function.

Write a C++ program for demonstrating the given situation for 10 products using array of objects. (CO2, CO3)

3. (a) Consider the class diagram given in Figure. The class Person derives information from both Faculty and Accountant classes which in turn derive information from Employee class. (CO4)

(i) Identify the type of inheritance. What problem is associated with this inheritance and how we can resolve this issue ?

(ii) Define all four classes and write a program to get and display the information contained in Person class.



(b) Consider the following declaration and answer the questions given below : (CO4)

class ABC

{

```
    int H :
  protected :
    int S;
  public :
    void input (int) ;
    void out() ;
};
class XYZ : private ABC
{
    int T;
  protected :
    int U;
  public :
    void indata(int, int);
    void outdata();
}
class PQR : public XYZ
{
    int M;
  public :
    void disp();
};
```

(i) Name the base class and derived class of the class XYZ.

(ii) Name the data member(s) that can be accessed from function disp().

(iii) Name the member function(s), which can be accessed from the objects of class PQR.

(iv) Is the member function out() accessible by the object of the class PQR ?

(c) How parameterized constructors of base classes are called using derived class in case of multiple inheritance ? Explain with the help of suitable example. (CO4)

4. (a) (i) Write short note on abstract class with example.

(ii) Describe properties of virtual function. Write a C++ program to show its application. (CO5)

(b) How is function overloading different from function overriding ? Explain with suitable example. (CO5)

(c) (i) What does this pointer point to ? Explain its application with the proper.

(ii) With the help of program, differentiate between static and non-static function of a class. (CO5)

5. (a) Explain different file handling operations available in C++. Write a program to copy the content of one file into another file. (CO6)

(b) Describe Exceptional handling mechanism in object-oriented programming. Write a C++ program to catch multiple exception using a single try block. (CO6)

(c) (i) Explain the concept behind generic programming ? Write a program to swap two numbers of different data types using function template.

(ii) What do you understand by Standard Template Library ? Explain its various components. (CO6)

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TCS–320

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

APPLICATION BASED PROGRAMMING IN PYTHON

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss looping statements in python along with flowchart and sample programs. (CO1)

(b) Design a menu driven Program in Python to perform the following tasks : (CO1)

(i) To check if an entered year is leap year or not

(ii) To calculate a GCD of two numbers

Support the program with appropriate functions for each of the above tasks.

P. T. O.

(c) Design a python program to perform following operations : (CO1)

(i) To count total number of Uppercase and Lowercase letters in the entered string data

(ii) To count the number of Vowels in the entered word

2. (a) Explain the following list methods with code snippet : (CO2)

(i) append

(ii) extend

(iii) count

(iv) insert

(v) remove

(b) Design a menu driven Program in Python for the following operations on Queue (List Implementation of Queue with maximum size MAX) : (CO2)

(i) Insert an Element on to Queue

(ii) Delete an Element from Queue

(iii) Demonstrate Overflow and Underflow situations on Queue

(iv) Display the status of Queue

(v) Exit

Support the program with appropriate functions for each of the above operation.

(c) Compare List, Tuple and Dictionary with examples. (CO2)

3. (a) Discuss the following file handling methods with code snippets : (CO3)
   (i) Read
   (ii) Append
   (iii) Create

   (b) Develop a menu driven Program in Python for the following Exceptions handling : (CO3)
   (i) ZeroDivisionError
   (ii) NameError

   (c) Discuss exception handling try, except, else and final block with example. (CO3)

4. (a) Discuss python regression following methods with code snippet : (CO4)
   (i) Search()
   (ii) Findall()
   (iii) Split()
   (iv) Sub()

   (b) Discuss any *ten* Metacharacters with example. (CO4)

   (c) Develop a python program to perform following operations : (CO4)
   (i) To extract domain name from a given e-mail id's
   (ii) To extract date from a given string

5. (a) Discuss method overloading and overriding with code snippets. (CO5, CO6)

   (b) Create a Python class called Student with USN, Name, Branch and Phone as variables within it. Develop a Python program to create n Student objects and print the USN, Name, Branch, and Phone of these objects with suitable headings. (CO5, CO6)

   (c) Explain Inheritance in python with examples. (CO5, CO6)

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TCS–321

B. TECH. (CE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

INTRODUCTION TO EMBEDDED SYSTEM

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) Differentiate between the following : (CO1)

(i) Harvard and Von Neumann

(ii) CISC and RISC

(b) With the help of a block diagram, explain the parts of basic structure of embedded system. (CO1)

(c) Explain the ROM and its types. (CO1)

2. (a) What is read modify write bus cycle ? (CO2)

P. T. O.

(b) Explain Bus Error and draw Pin diagram of 8051 Microcontroller.(CO2)

(c) With the help of a diagram, explain he internal data memory organization by defining memory used for every particular block of data memory. (CO2)

3. (a) Explain the following : (CO3)

   (i) 8051 flag bits and PSW register with the help of a diagram.

   (ii) Register banks and its selection through PSW and Stack

   (b) Write short notes on the following : (CO3)

   (i) Assemblers

   (ii) Simulators

   (c) What are special function registers ? (CO3)

4. (a) Explain the following : (CO3, CO4)

   (i) Timers and Counters in 8051

   (ii) Interrupts and interrupt priority in 8051

   (b) With the help of an example, explain what are various Instruction Groups. (CO3, CO4)

   (c) With the help of an example, explain what are various Addressing Modes of 8051. (CO3, CO4)

5. (a) Write a program for addition of two 16-bit numbers, the content is stored in 45H, 46H and 55H, 56H.

   Write a program for BCD addition of two 8-bit numbers the content is stored in 23H and 24H. (CO4)

   (b) With the help of a diagram, explain the interfacing between 8255 and 8051. (CO5)

   (c) Define basics of serial communication in 8051. (CO6)

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TCS–331

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

FUNDAMENTAL OF IOT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Particle is an IoT platform that combines software, hardware, and connectivity as an integrated solution. This IoT service provider helps you to ensure that IoT products are reliable, scalable, and secure. Discover the applications of the Internet of things with the perspective of Particle. (CO1)

   (b) Softeq is a global full-stack IoT development firm that designs and develop IoT services, custom software, desktop apps, mobile apps, big data, and machine learning-powered solutions. Apply the key technologies of the Internet of Things and create a framework from the perspective of Softeq. (CO1)

(c) Konstant Infosolutions is a widely known mobile app development company delivering best-in-class IoT solutions. This firm uses current technologies and tools for IoT solutions to provide a seamlessly connected experience to the customers. Integrate the concept of Machine learning with the Internet of Things and generate a framework from the perspective of Konstant Infosolutions. (CO2)

2. (a) Mobiloitte is an IoT service provider that helps you to make your business smart with its world-class, cutting-edge IoT application development solutions and services. Classify the IoT sensors and evaluate Mobiloitte's loT framework on the bases of the sensors used. (CO2)

(b) Decathlon is one of the world's significant retailers specializing in sportswear, shoes and other merchandise. Decathlon's main objective was to make sure that the product is available in the store if the customer is willing to purchase. Decathlon did not want to disappoint any customer. In simple words, they do not want to lose the sale. Therefore, they utilized RFID technology. Justify the role of RFID along with its components from the perspective of Decathlon. (CO3)

(c) MobiDev is an IoT service provider firm that helps you to create complex business-driven solutions. It focuses on innovation and transparency of actions, guaranteed product delivery, and continuous evolution. The company mainly focuses on Internet of Things, Augmented Reality and Data Science. Evaluate an IoT product based on its essential characteristics from the perspective of MobiDev. (CO3)

3. (a) AT&T Inc., the cellular service provider based in Dallas TX, is also working to provide IoT solutions using their own AT&T IoT platform. They have set up a device-based DataFlow infrastructure that allows designers to rapidly develop customized solutions to solve tricky business challenges. Build a Wireless sensor network topology for AT&T Inc. and explore its characteristics. (CO3)

   (b) DCSL Software is one of the UK's leading loT development companies. This firm helps you design intelligent, cost-effective, and intuitive IoT solutions, web applications, desktop applications, and mobile apps to streamline your business processes. Design an IoT protocol stack from the perspective of DCSL Software. (CO4)

   (c) iTechArt Group is a custom software development company founded in 2002 that offers secure IoT solutions to startups and tech companies. iTechArt builds IoT apps, gateways, data analytics, and third-party integrations and has more than 1,800 staff. Compare and contrast various Hientity management models, and identify which model is optimum for iTechArt Group. (CO4)

4. (a) Telecommunications company Huawei offers IoT solutions like Connected Car, Public Utilities, and Predictive Maintenance. It provides services like IoT Platform, Machine Learning Service, and Cloud Stream Service. Interpret the data transmission and security challenges from the perspective of Huawei. (CO4)

(b) ARM provides the IoT industry with a Device-to-Data Platform for connectivity management, device management, and data management. It also provides device products like Mbed OS, SoC Solutions, and Kigen SIM Solutions. Justify the role of Software agents in ARM. (CO4)

(c) German automation company Siemens offers an intelligent gateway for IoT solutions. Identify and manage the resources used in IoT from the perspective of Siemens. (CO5)

5. (a) Vates' IoT Engineers and Systems Integrators employ high-end prototyping services to ensure its customer's product is market-ready. Construct a Security model for loT from the perspective of Vates. (CO5)

(b) The purpose of the Smart Cities Mission is to drive economic growth and improve the quality of life of people by enabling local area development and harnessing technology, especially technology that leads to Smart Identify and analyze the smart city applications. (CO6)

(c) A smart home is a residence that uses internet-connected devices to enable the remote monitoring and management of appliances and systems, such as lighting and heating. Design the use cases and misuse cases for smart home. (CO6)

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Roll No. ....

TCS–342

B. TECH. (CSE-AI & DS) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

INTRODUCTION TO STATISTICAL DATA SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Develop IQR and Z-score algorithms to find outliers. (CO1)

(b) The weekly wages of 500 workers are normally distributed around a mean of \$500 with a standard deviation of \$40. Estimate the number of workers, whose weekly wages will be : (CO1)

(i) between \$400 and \$600

(ii) less than \$400

(iii) more than \$600

P. T. O.

Given that :

$$P(0 < Z < 2) = 0.4772$$

$$P(Z < 0.4) = 0.6554$$

$$P(Z < -0.6) = 0.2743$$

(c) Discuss the modern data analysis methods, process and types. (CO1)

2. (a) Find the population covariance matrix for the following table : (CO2)

Score	Age
68	29
60	26
58	30
40	35

(b) Elaborate sample covariance, sample covariance matrix and correlation. (CO2)

(c) The heights (in feet) of the different person are as follows : (CO2)

1.2, 2.3, 4.9, 5.1, 5.2, 5.8, 5.9, 6.2, 6.5, 7.1, 24.5, 33.2, 40.4

Find heights outlier by applying Z-score method.

3. (a) A survey was conducted in your city. Given is the following sample data containing a person's age and their corresponding income. Find out whether the increase in age has an effect on income using the Pearson's correlation coefficient formula : (CO3)

Age	Income (₹)
25	30,000
30	44,000
36	52,000
43	7,000

(b) The following data shows the sales (in million dollars) of a company : (CO3)

x	y
2015	12
2016	19
2017	29
2018	37
2019	45

Can you estimate the sales in the year 2020 using the regression line ?

(c) Discuss the importance of data cleaning, correlation and outliers' detections in data analytics. (CO3)

4. (a) Write short notes on the following : (CO4)

(i) The hypothesis-testing

(ii) Z-score

(iii) Central Limit Theorem

(b) Classify the test data "*A very close game*" into right class as sports or not sports using Naïve Bayes classification : (CO4)

ID	Keywords	Review class
1	A great game good game	Sports
2	This election game was over	Not Sports
3	A very closed match	Sports
4	A clean and forgettable game	Sports
5	That was very closed election	Not Sports
6	Exam was very tough	Not Sports

(c) Why do we need data exploration and preparation during data analytics ? Elaborate in detail. (CO4)

5. (a) Three types of fertilizers are used on three groups of plants for 5 weeks. We want to check if there is a difference in the mean growth of each group. Calculate the ANOVA coefficient : (CO5, CO6)

Fertilizer 1	Fertilizer 2	Fertilizer 3
6	8	13
8	12	9
4	9	11
5	11	8
3	6	7
4	8	12

(b) Calculate the Chi-square value for the following data of incidences of water-borne diseases in three tropical regions : (CO5, CO6)

	India	Equador	South America	Total
Typhoid	31	14	45	90
Cholera	2	5	53	60
Diarrhoea	53	45	2	100
Total	86	64	100	250

(c) Discuss Parametric testing with examples. (CO5, CO6)

100

Roll No. ....

TCS–351

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

FUNDAMENTAL OF CLOUD COMPUTING AND BIG DATA

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) With abundance of data over various Social Media sites, there exist various gaps in traditional systems to work upon its data analysis. A quick solution to the can be Big Data Analytics Tool. Review various available big data tools present market and rate them according to their work profiles. (CO2)

(b) Distinguish between the following : (CO2)

(i) Supervised and Unsupervised Learning

(ii) Association Rule Mining and Clustering

P. T. O.

(c) Sophie working as a data analyst in DataLuke company. She received a dataset from her client AccuLogic with attributes like employee name, age, gender, working hours, payment per hour and salary. AccuLogic is planning to automate its employee's salary hike according to the mentioned dataset. Mention the machine learning algorithms that can help Sophie to develop a solution. (CO2)

2. (a) Evaluate the type of data in below : (CO1)

   (i) Weather data stored in Relational Database

   (ii) Jerry mails to John

   (iii) Data received after browsing

   (iv) Personal Data stored in XML

   (b) Discuss Primary Data Mining goals in real world. (CO1)

   (c) HTM, a popular superstore chain is planning to expand its business online to increase its customer base and reachability. Access the capability, advantages, impact of Web-3.0 for developing a solution for this company. (CO1)

3. (a) Amphilo, an online restaurant generates at least 5GB of data in a single day. The company wishes to provide vouchers to its old customers and attract new customers. How can data analysis techniques help Amphilo for achieving its wish ? (CO4)

   (b) Evaluate the use of Unsupervised Learning Algorithms in case of : (CO4)

   (i) Development of Recommendation Engines

   (ii) Anomaly Detection System

(c) "Cloud Storage delivers on demand, just in time capacity and low cost storage." Justify this statement by describing its requirement, types, work and benefits. (CO4)

4. (a) FanDuel Group, the leading sports gaming company in the United States, has accelerated its business expansion and improved its user experience by standardizing on Amazon Web Services (AWS) across both its cloud and on premises operations using AWS Outposts. Evaluate the working of Cloud Computing and the impact of networks from perspective of FanDuel Group. (CO3)

(b) Discuss the role of SLA in an IT company. (CO3)

(c) "Big Data and Cloud Computing are not competitors, but collaborators for modern world analytics."—Yes/No. Explain with the help of a real time example. (CO3)

5. (a) Validate the need of Cyber Security in current paradigm. Discuss various domains wherein it acts as backbone. (CO5)

(b) Scalablity helps handling the growing workload with high performance. Discuss and compare its types. (CO5)

(c) Xingo, a telecom company is trying to locate a fraudster using its network. The company gets data from varied sources such as IP database, Tower database, Adhar data of its users, ID Proof data, etc. and requires integrating it in one schema so that the fraudster could be located. Mention various data integration tools that can be helpful for the company. (CO5)

480

Roll No. ....

TCS–372

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

MATHEMATICS FOR AI AND MACHINE LEARNING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) Define qualitative data and quantitative data with proper examples. (CO1)

(b) Elucidate the span of vectors with its properties. Calculate the span of two dependent vectors (1, 1) and (2, 4). (CO1)

(c) Define convex sets and convex functions with proper schematic diagram. What do you understand by penalty function methods ? Define its pros and cons. (CO1)

P. T. O.

2. (a) Determine the eigen values for the following matrix : (CO2)

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 1 \end{bmatrix}.$$

(b) Find the eigen vector for the following matrix : (CO2)

$$A = \begin{bmatrix} 1 & -5 \\ -3 & -8 \end{bmatrix}.$$

(c) Define a Jacobian matrix with suitable examples. Let $x(u, v) = u^2 - v^2$, $y(u, v) = 2uv$. Determine the Jacobian J (u, v). (CO2)

3. (a) Elucidate the Gram-Schmidt orthonormalization process with the purpose of it in terms of application in artificial intelligence and machine learning. (CO3)

(b) Assume the following matrix. Apply the Gram-Schmidt process to $\{V_1, V_2, V_3\}$: (CO3)

$$V_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, V_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix}, V_3 = \begin{bmatrix} 0 \\ -1 \\ 2 \\ 1 \end{bmatrix}.$$

(c) Define the Hessian matrix with proper formulation. Why is the Hessian matrix important in machine learning ? Find Hessian (H_g) and the discriminant (D_g) for the following equation : (CO3)

$$g(x, y) = 2x^2 + 3y^2 + 5xy^2.$$

4. (a) Define the concept of partial derivatives with all its four rules. (CO4)

(b) Find the partial derivative of $f(x, y) = x^2y + \sin x + \cos y$. (CO4)

(c) With proper mathematical equations, describe the Newton-Raphson method. Find the root of the equation $x^2 - 4x - 7 = 0$ near $x = 5$ to the nearest thousandth by using Newton-Raphson method. (CO4)

5. (a) Prove that $C = \{(x_1, x_2) : 2x_1 + 3x_2 = 7\} \subset R^2$ is a convex set.

(CO5, CO6)

(b) Elucidate the Nelder-Mead method with proper examples. (CO5, CO6)

(c) Define the optimization and its formulation process. Write a pseudo code for PSO optimization method with proper flow diagram.

(CO5, CO6)

80

Roll No. ....

TCS–391

B. TECH. (CSE/CE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

FUNDAMENTAL OF CYBER SECURITY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) In an ongoing network communication there are the chances that the existing adversary may modify the exchanged messages. Explain any security mechanism, which can be used to secure such kind of communication. Write down all steps clearly. (CO6)

(b) Write down the summary of the following : (CO6)

Birthday attack, Cyber vandalism and Hacktivism.

(c) How would you compare Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) attacks ? Which one is more devastating ? (CO6)

P. T. O.

2. (a) How would you compare e-mail phishing and spear phishing attacks ? Which one is more devastating for the ongoing online operations of an organization ? (CO2)

   (b) Write down a python program to read data from a file and then write that data in another created file. Also measure the computation time of your program. (CO2)

   (c) Explain the mechanism of TCP 3-way handshake using some figures. Why is the connection establishment required ? (CO2)

3. (a) Investigate the working mechanism of reflected cross site scripting (XSS) attack. What are the different mitigation methods of XSS attacks ? (CO4)

   (b) How would you compare in-band and inferential (blind) SQLi attacks ? What are the different solutions available for SQLi attacks ? (CO4)

   (c) Demonstrate the mechanism of the following NMAP scans using their syntax : (CO4)

   Version scan, Aggressive scan and Decoy scan.

4. (a) Write down the summary of the following malware programs : (CO3)

   Ransomware, Trojan horse and Worm.

   (b) Why do we need access control mechanism ? How would you compare role based access control (RBAC) and mandatory access control (MAC) mechanisms ? (CO3)

   (c) Explain the uses of the following Linux commands : (CO3)

   man, chmod, cat, touch and sudo.

5. (a) Describe the following cyber security key terms : (CO1)

vulnerability, threat, threat agent and attack

(b) Demonstrate the working mechanisms of the following spoofing attacks : (CO1)

ARP spoofing attack and IP spoofing attack.

(c) What is anonymous browsing ? What are the different ways through which anonymous browsing can be performed ? (CO1)

260

Roll No. ....

TCS–301

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

LOGIC DESIGN

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Minimize the following Boolean expression by using tabular method. Implement minimized expression by using EX-OR gate only. (CO1 and CO2)

$$Y(A, B, C, D) = \Sigma\, m\,(5, 7, 8, 10, 13, 15) + \Sigma\, d\,(0, 1, 2, 3)$$

(b) If a Hamming code received as 1 1 1 1 1 0 0 1 0 1 1 1. Find whether the received code is correct or not for even parity. If wrong find the error and correct it. (CO1 and CO2)

P. T. O.

(c) Implement 2 bit magnitude comparator by using suitable decoder only. (CO1 and CO2)

2. (a) What are the differences between PAL and PLA ? Implement full adder by using PAL and PLA. (CO2)

(b) Draw full adder by using half adder and suitable logic gate. Also implement the following functions by using half adder only. (CO2)

$F1 = ABC' + A'C + B'C$.

$F2 = A'BC + AB'C$

(c) Design BCD to excess-3 code converter. (CO2)

3. (a) Implement 2 input universal gates and basic gates using 2 to 1 multiplexer. (CO2 and CO3)

(b) What do you mean by race around condition ? Convert a T Flip-flop into SR Flip-flop. (CO2 and CO3)

(c) Draw any five flip flop circuits which give toggle condition. (CO2 and CO3)

4. (a) Draw and explain universal shift register. (CO4)

(b) Design a synchronous counter using T flip flop for count the sequence 4, 6, 7, 3, 1, 4 .... Avoid lock out condition. (CO4)

(c) Draw and explain operation of 4 bit creeping counter. Also determine total number of unused states for 4 bit creeping counter. (CO4)

5. (a) An asynchronous sequential logic circuit is described by the following excitation and output function : (CO5 and C06)

$$Y = X1X2' + (X1 + X2)\,Y$$

$$Z = Y$$

(i) Draw the logic diagram of circuit.

(ii) Derive the transition table and output map.

(b) Design a clocked sequential circuit (by using D flip flop) which detects 3 or more consecutives 1's. Also draw ASM chart. (CO5 and C06)

(c) Design a ripple counter using T flip flop whose input frequency is 2 MHz and its output frequency is 200 KHz. (CO5 and C06)